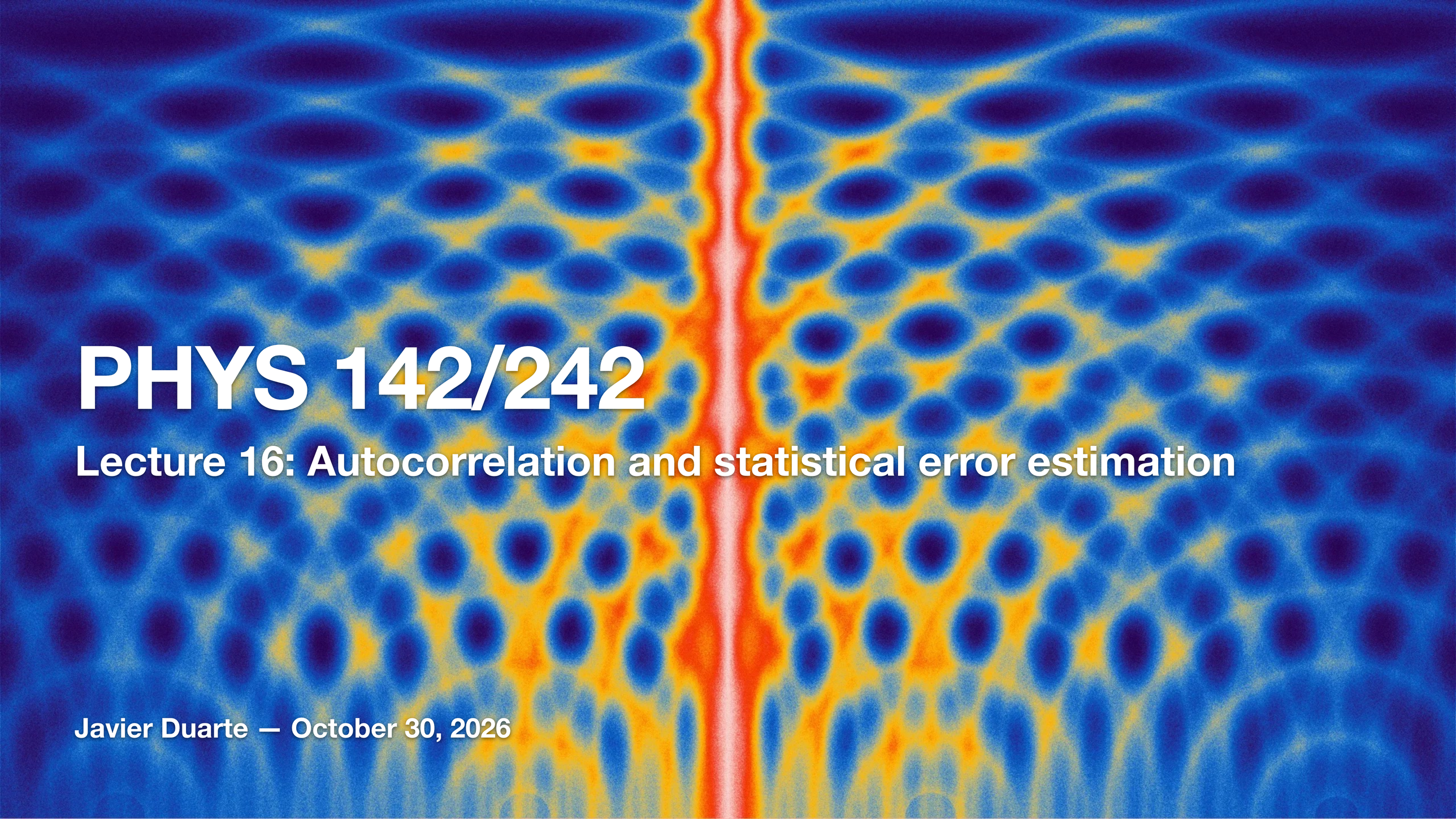


PHYS 142/242

Lecture 16: Autocorrelation and statistical error estimation

Javier Duarte — October 30, 2026

Markov process and chain

A ***Markov process*** is a set of probabilistic rules that tell us how to choose a new state of the system, based only on the system's current state. If the system is currently in state n then the probability of choosing state m on the next step is given by the “transition probability” $P(m|n)$.

By repeatedly applying the Markov process, we move the system through a random sequence of states, $\{n^{(0)}, n^{(1)}, n^{(2)}, \ldots\}$ where $n^{(k)}$ denotes the state on step k.

This random sequence is called a ***Markov chain***.

Because the system must transition to some state,

$$\sum_m P(m|n) = 1 \quad \text{for all } n \in \{0,1,\ldots\}\,.$$

Metropolis algorithm

Design a Markov process to match a given stationary distribution π_n based on **detailed balance**

$$P(n \mid m)\, \pi_m = P(m \mid n)\, \pi_n \quad \text{for all } m, n$$

1. On step k, the system is in state n. Randomly choose a candidate state m with **proposal** probability $q(m \mid n) = q(n \mid m)$

2. Accept the candidate with probability $A(m \mid n) = \min\left(1, \frac{\pi_m}{\pi_n}\right)$

3. If the candidate is accepted, the state in step $k+1$ is m. Otherwise, the state in step $k+1$ remains n

4. Repeat

Metropolis-Hastings algorithm

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Practical considerations for MCMC

- Several practical considerations
  - Acceptance rate: the rate at which we accept proposals is important. We want this to be relatively high (to reduce total running time), but not always 1 (which may mean we're not sufficiently sampling the full space). Typical values are 15% to 50%
  - Burn in
  - Error and (integrated) autocorrelation time

Burn-in

"Burn-in": common to throw out the first few states of a Markov chain, maybe the first 100 or the first 1000

The idea is to get rid of "transient behavior" connected to an improbable initial configuration

Alternative: pick an initial state near equilibrium (i.e. not unlikely)

Autoregressive series

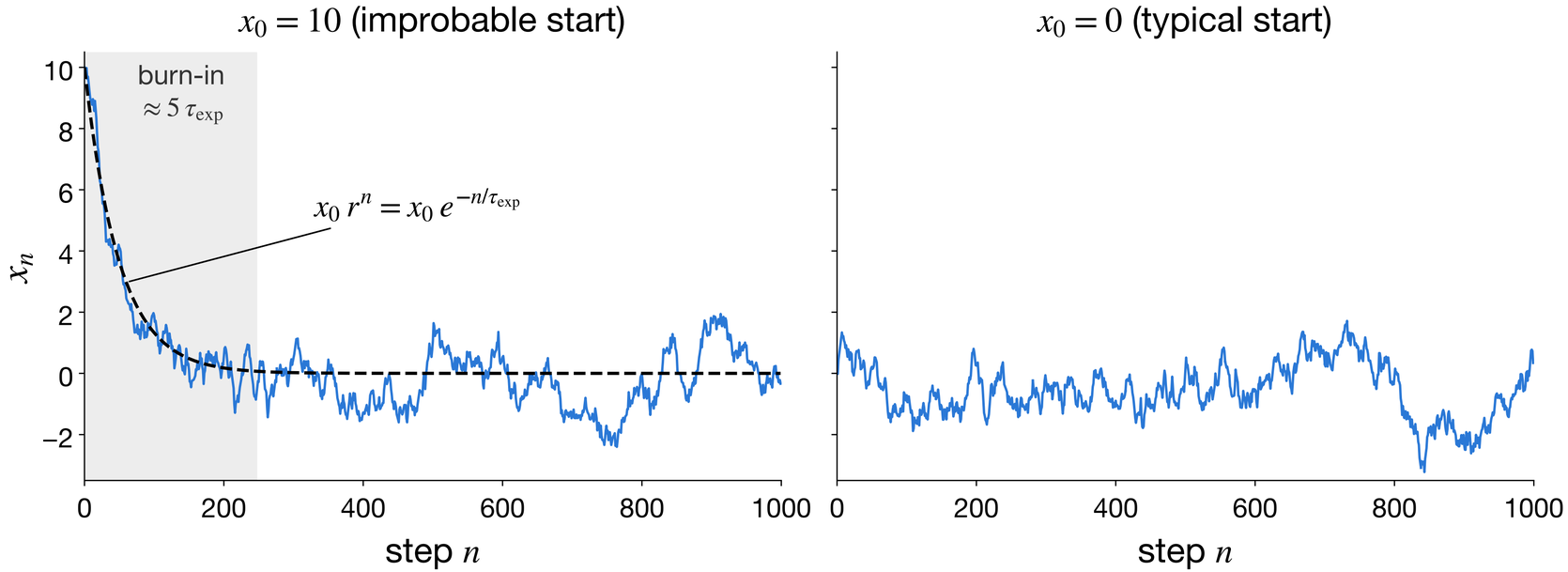
$$x_{n+1} = rx_n + e_n$$

with $r = 0.98$ and

$$e_n \sim \mathcal{N}(0, 1 - r^2)$$

starting from $x_0 = 10$ (left)

and $x_0 = 0$ (right)



$$\rho(t) = r^t = e^{-t/\tau_{\text{exp}}}, \quad \tau_{\text{exp}} = -1/\ln r \approx 50$$

MC error

Our goal is to estimate the expectation value of some physical observable f.

Using traditional MC sampling, the estimate is given by the average over N samples

$$\bar{f} = \frac{1}{N}\sum_{i=1}^{N} f_i$$

If these samples are independent, then the variance on this estimator is

$$\sigma^2_{\bar{f}} = \frac{1}{N}\mathrm{Var}[f_i] = \frac{1}{N}\left\langle (f_i - \langle f\rangle)^2 \right\rangle = \frac{1}{N}\left(\langle f^2\rangle - \langle f\rangle^2\right)$$

and the error decreases as $1/\sqrt{N}$ as we generate more samples.

MCMC error

However, in MCMC, the samples are not independent! What is the **variance**?

Autocorrelation function

Useful to define the autocorrelation function of the observable f as

$$c_f(t) = \left\langle (f_i - \langle f \rangle)(f_{i+t} - \langle f \rangle) \right\rangle \approx \hat{c}_f(t) = \frac{1}{N} \sum_{i=1}^{N-t} (f_i - \bar{f})(f_{i+t} - \bar{f})$$

Note variance is a special case $c_f(0) = \mathrm{Var}[f_i]$ and symmetry $c_f(-t) = c_f(t)$

We can also define the normalized autocorrelation (like a Pearson correlation coefficient)

$$\rho_f(t) = c_f(t)/c_f(0)$$

It's probably more appropriate to define $c_f(t)$ as "autocovariance," but in practice you have to figure out what the literature means from context

MCMC error

Stationarity: after burn-in, $c_f(t)$ depends only on the lag $t = j - i$, not on i

However, in MCMC, the samples are not independent! What is the **variance**?

$$\sigma_{\bar{f}}^2 = \mathrm{Var}\left[\frac{1}{N}\sum_{i=1}^{N} f_i\right] = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N} \underbrace{\left\langle (f_i - \langle f\rangle)\,(f_j - \langle f\rangle)\right\rangle}_{c_f(j-i)\ \text{(stationarity)}}$$

$$= \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N} c_f(j-i) = \frac{1}{N^2}\sum_{t=-(N-1)}^{N-1} (N - |t|)\, c_f(t)$$

$$= \frac{c_f(0)}{N}\sum_{t=-(N-1)}^{N-1}\left(1 - \frac{|t|}{N}\right)\rho_f(t)$$

$$\approx \frac{\mathrm{Var}[f_i]}{N}\sum_{t=-\infty}^{\infty}\rho_f(t) \equiv \frac{\tau_f}{N}\mathrm{Var}[f_i] \qquad (N \gg \tau_f)$$

$N = 4$: entry (i, j) is $c_f(j - i)$

$$\begin{pmatrix} c_f(0) & c_f(1) & c_f(2) & c_f(3) \\ c_f(-1) & c_f(0) & c_f(1) & c_f(2) \\ c_f(-2) & c_f(-1) & c_f(0) & c_f(1) \\ c_f(-3) & c_f(-2) & c_f(-1) & c_f(0) \end{pmatrix}$$

where τ_f is the ***integrated autocorrelation time*** for the observable f

Samples about τ_f apart are roughly independent

Integrated autocorrelation time

The integrated autocorrelation time is then given by

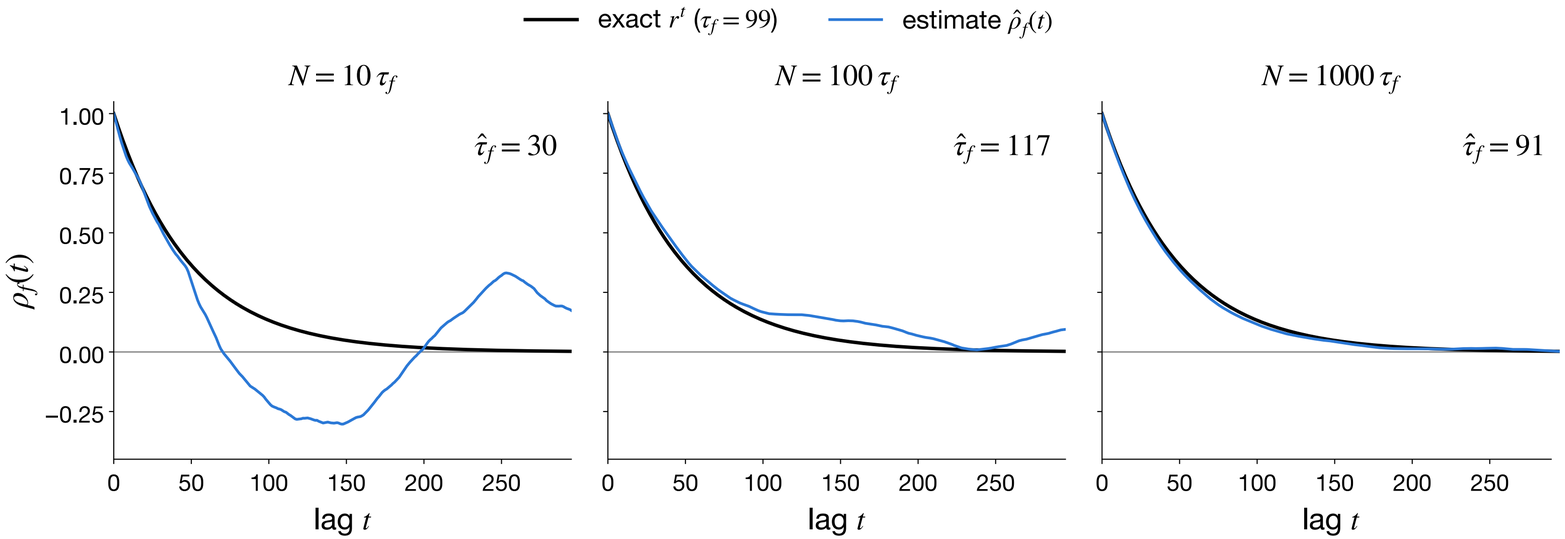
$$\tau_f = \sum_{t=-\infty}^{\infty} \rho_f(t) = 1 + 2 \sum_{t=1}^{\infty} \rho_f(t)$$

Trying to approximate this directly is difficult. At longer lags, $\hat{\rho}_f(t) = \hat{c}_f(t)/\hat{c}_f(0)$ starts to contain more noise than signal and summing all the way out to N gives $\hat{\tau}_f = 0$ exactly. Instead we want to cut off the sum at some $M \ll N$

$$\hat{\tau}_f(M) = 1 + 2 \sum_{t=1}^{M} \hat{\rho}_f(t), \qquad \text{smallest } M \text{ with } M \geq 5\,\hat{\tau}_f(M)$$

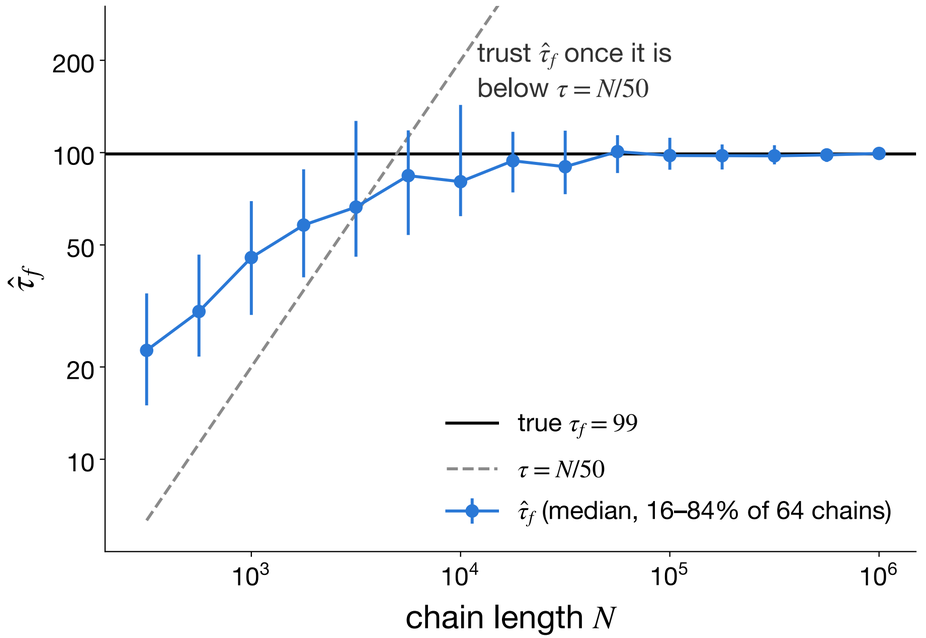
Estimating τ_f in practice

Estimating integrated autocorrelation time: https://emcee.readthedocs.io/en/stable/tutorials/autocorr/



Short chains underestimate τ_f, and therefore the error bar; Trust $\hat{\tau}_f$ only once $N \gtrsim 50\,\hat{\tau}_f$

Function: emcee.autocorr.integrated_time(x)



Blocking: error bars without $\rho(t)$

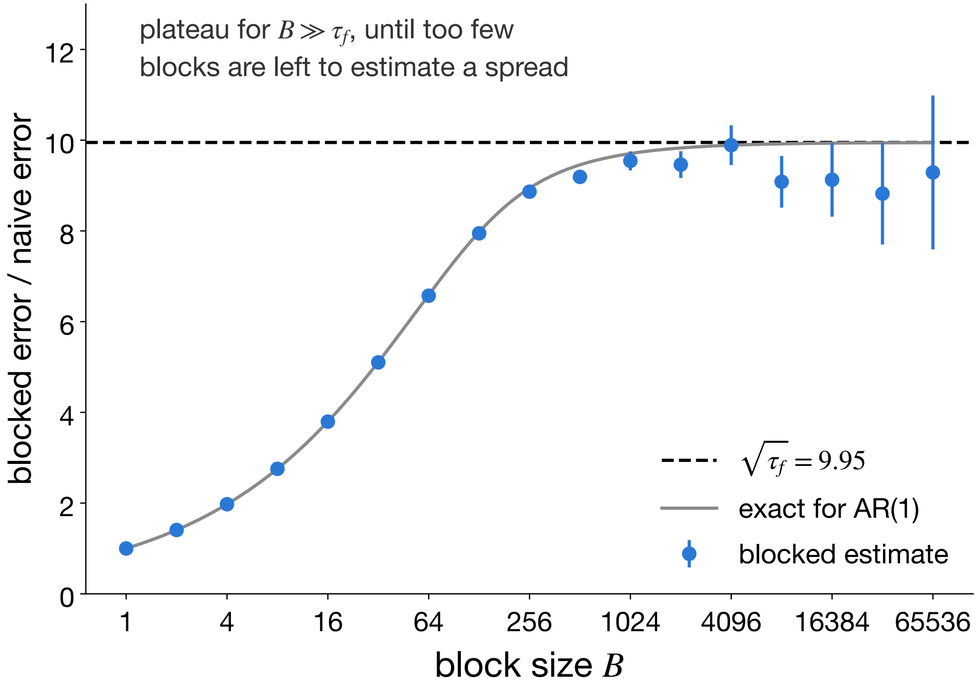
1. Split the chain into $N_B = N/B$ blocks of B steps and average each block:

$$F_b = \frac{1}{B} \sum_{i=(b-1)B+1}^{bB} f_i, \qquad b = 1, \ldots, N_B = \frac{N}{B}$$

2. Treat the block means as independent samples and apply $(B \gg \tau_f)$

$$\sigma^2_{\bar{f}}(B) = \frac{1}{N_B(N_B-1)} \sum_{b=1}^{N_B} \left(F_b - \bar{f}\right)^2 \approx \frac{\tau_f}{N} \mathrm{Var}[f_i]$$

3. Double B over and over. The error grows and then levels off; the level part (the plateau) is the error.



Once $B \gg \tau_f$, neighboring blocks are nearly uncorrelated and each F_b has variance $\approx \tau_f \mathrm{Var}[f_i]/B$. So $\sigma^2_{\bar{f}}(B) \to \tau_f \mathrm{Var}[f_i]/N$

$B = 1$ gives the naive error. So on the plateau, $\sigma_{\bar{f}}(B)/\sigma_{\bar{f}}(1) \approx \sqrt{\tau_f}$. Naive error is 10× too small!

If there's no plateau, the chain is too short. Same idea as $N \gtrsim 50\hat{\tau}_f$